

Final Project: Basic System Design for a Small Business Ordering System

Estimated time: 60 minutes

Welcome to the final project!

Crusty Muse is a small bakery that serves local customers. Currently, customers can only order in person or by phone.

In this final project, you will apply basic systems architecture concepts to design a simple online ordering system for the bakery.

Objectives

In the final project, you will:

- Identify the stakeholders
- Apply system fundamentals (input/process/output)
- Select the appropriate SDLC model based on project characteristics
- Choose the architecture pattern appropriate for business size and complexity
- Create a basic system flow
- Document your decisions

Business scenario: Crusty Muse Bakery

Crusty Muse Bakery wants a simple online ordering system for the bakery. On gathering the requirements, here's what you learn:

What the system needs to do:

- Show available products (cakes, bread, and pastries) with prices
- Allow customers to add items to a cart and place orders
- Allow the staff to view new orders and update their status
- Send confirmation emails to customers

Constraints:

- Small budget
- Simple to use and maintain
- Capable of handling about 20-30 orders per day
- Should work on mobile phones

Software used for this project

We recommend using Microsoft's "Word for the Web," one of the most used online word-processing software applications worldwide. To access this application, you will need to sign up for a free Microsoft 365 account. For instructions on how to sign up for a free account, [click here](#). For step-by-step guidance on getting started with Word Online, [click here](#).

You can alternatively use any word-processing software or text editor for this exercise.

Template to use

If you're using Microsoft Word, you can download the [Final Project template.docx](#). Right-click the link and download the file. You can use the template to perform the tasks outlined in the project.

Project instructions

Task 1: Conduct basic system analysis

Memory refresher from videos: *"A system is a set of interrelated components working together towards a common goal. System characteristics include components or subsystems, interrelatedness, boundary, input and output, process, and goal orientation. Every system is made up of parts that are connected; changes in one part affect others."*

A: Identify key stakeholders

Instructions: Select four stakeholders who will interact with this online ordering system. Strictly choose from the options below. Also specify their role in the online ordering system.

Available options:

- Customers (people buying baked goods)
- Bakery staff (taking and preparing orders)
- Bakery manager/owner
- Delivery personnel
- Payment processor (like PayPal)
- Website hosting company

Example:

Stakeholder: Customer

Role: Select the goods they want to buy and place the order

Use this template to record the stakeholder and their role in the online ordering system.

Stakeholder	Role

B. Identify the system components

Memory refresher from videos: "Systems take inputs, process them, and produce outputs. Input refers to resources such as data, materials, or energy required for the system to function. Process refers to the transformation that occurs within the system. Output is the result or product of the system."

Instructions: Given a use case, provide the input (what goes into the system), process (what the system does with the input), and output (what comes out of the system) for it.

Here's an **example:**

Use case: Customer order

Input: Order details

Process: Check inventory

Output: Order confirmed

Similarly, complete the table with one input, process, and output for "Status confirmation email to the customer" in the provided template.

Use case	Input (what goes into the system)	Process (what the system does with the input)	Output (what comes out of the system)
Status confirmation email to the customer			

Task 2: Select an SDLC model

Memory Refresher from Videos: "The Waterfall model is a linear approach where each phase is completed before the next begins. It's best suited for projects with well-defined and unchanging requirements. The Agile model is an iterative approach that delivers small parts of the system in sprints. It promotes flexibility for changing requirements."

Instructions: Choose the model that is appropriate for developing the online ordering systems for Crusty Muse. Provide the reason for your choice.

- Option 1: Waterfall
- Option 2: Agile

Choice of SDLC model	
Reason	

Task 3: Select the architecture pattern

Memory refresher from videos: "The 2-tier architecture, also called client-server, has an interface on the client that makes requests to a server. The 3-tier architecture has three layers: presentation tier (user interface), middle tier (business logic), and data tier (where data is stored). Each tier only communicates with tiers directly above and below it."

Instructions: Select one of the two architecture patterns and provide the reason for your choice.

Option 1: 2-Tier (Client-Server)

- Client: Customer's web browser
- Server: Handles everything (showing products, processing orders, storing data)
- Simpler to build and maintain
- Good for small businesses

Option 2: 3-Tier

- Presentation: Web pages customers see
- Business logic: Processes orders and calculations
- Data: Stores products, orders, and customer info
- More organized but more complex

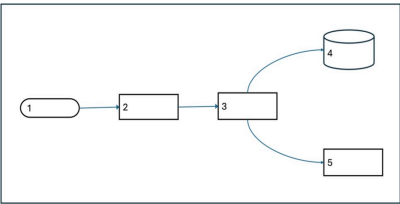
Hint: Think about what's simplest for a small bakery with a small budget.

Choice of architecture pattern	
Reason	

Task 4: Complete a basic system diagram

Memory refresher from videos: "In a client-server architecture, the interface resides on the client machine and makes requests to a server for data or services. The server hosts, delivers, and manages most of the resources and services delivered to the client."

Instructions: Fill in the missing parts of this basic system flow. Use only these components: Website, Server, Database, Staff Dashboard, and Customer. Complete the table in the template with the correct answer.



Complete the table in the template with the correct answer.

	Component
1	
2	
3	
4	
5	

Task 5: Document your progress in the project

Memory refresher from videos: "Systems thinking focuses on how things are connected and work together rather than breaking them down into isolated parts. A system exists to achieve a specific purpose or goal."

Instructions: Complete the project summary in the template.

Project summary	
Business problem	
System Goal	
Technical choices	
SDLC Model	
Reason for the choice	
Architecture pattern	
Reason for the choice	

Complete all the tasks from 1 to 5 in the template provided and save the Word document. You can then submit the Word document for AI-enabled evaluation. Please note, while you receive an evaluation, the points are not counted toward your final grade.



Skills Network